ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

CELL EXTRACTION FROM STAINING IMAGES

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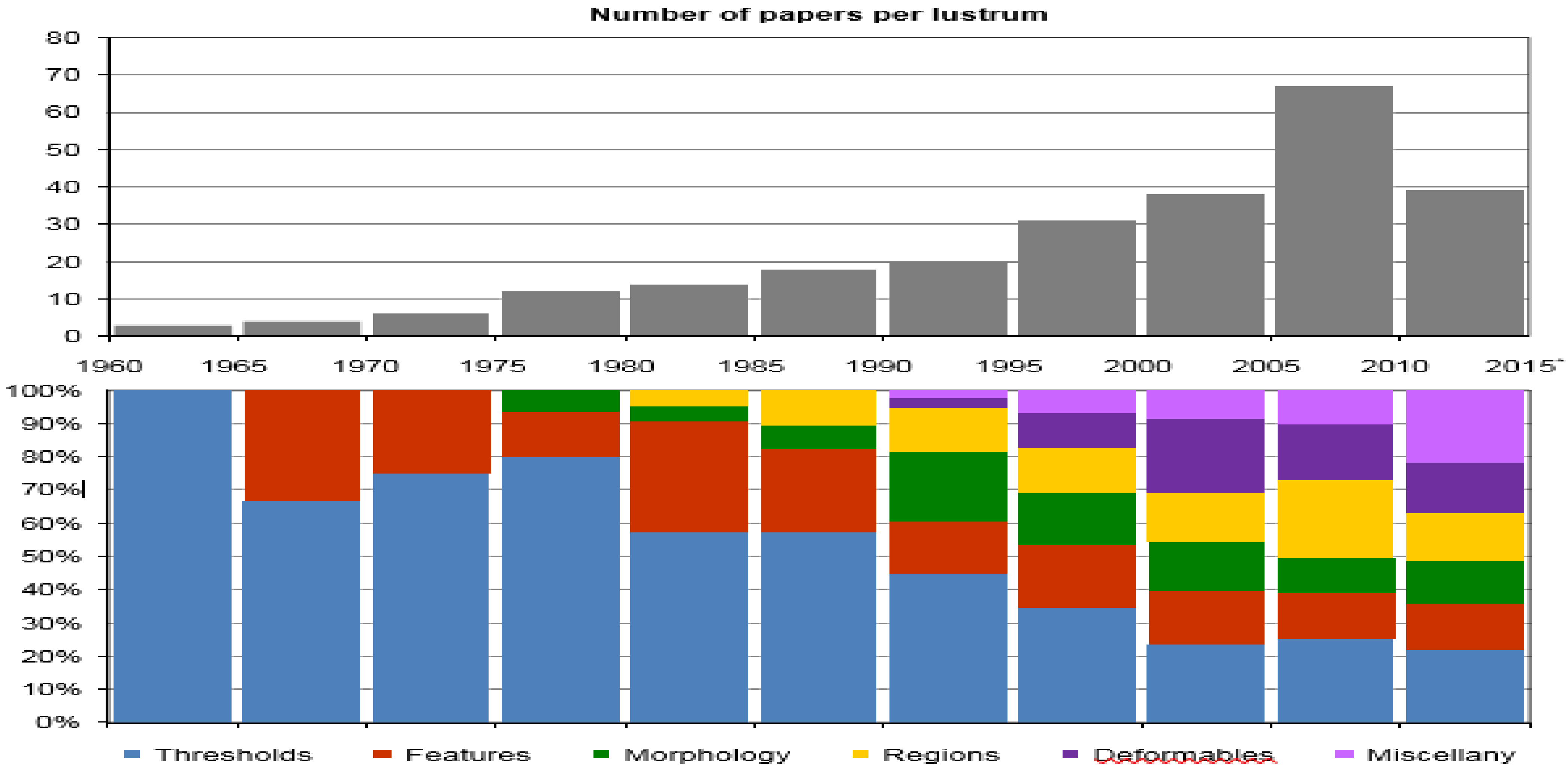
Introduction and Motivation

- Use of computers for cell analysis dates back more than 50 years.

- Nowadays, basic structure is preprocessing, segmentation and postprocessing

- Main issues: “…large variability (different microscopes, stains, cell types, cell densities) and complexity of the data (acquired at multiple wavelengths, using multiple microscopes, and containing large numbers of cells)” (Meijering 2).

- To create an all-compassing automated algorithm, we first need a special preprocessing method.



Count of published papers and the distribution of the methods of cell extraction (Meijering 6)

Descriptions of the Methods

- **K-means Clustering:** Data is assigned to random initialized centers with Euclidian distance, centers are recalculated for each cluster, then algorithm recursively applied until a point of stability given by user.

- **Watershed:** “Power watershed“ described by Couprie et al., combining random walker, graph cuts and optimal spanning forest algorithm for watershed.

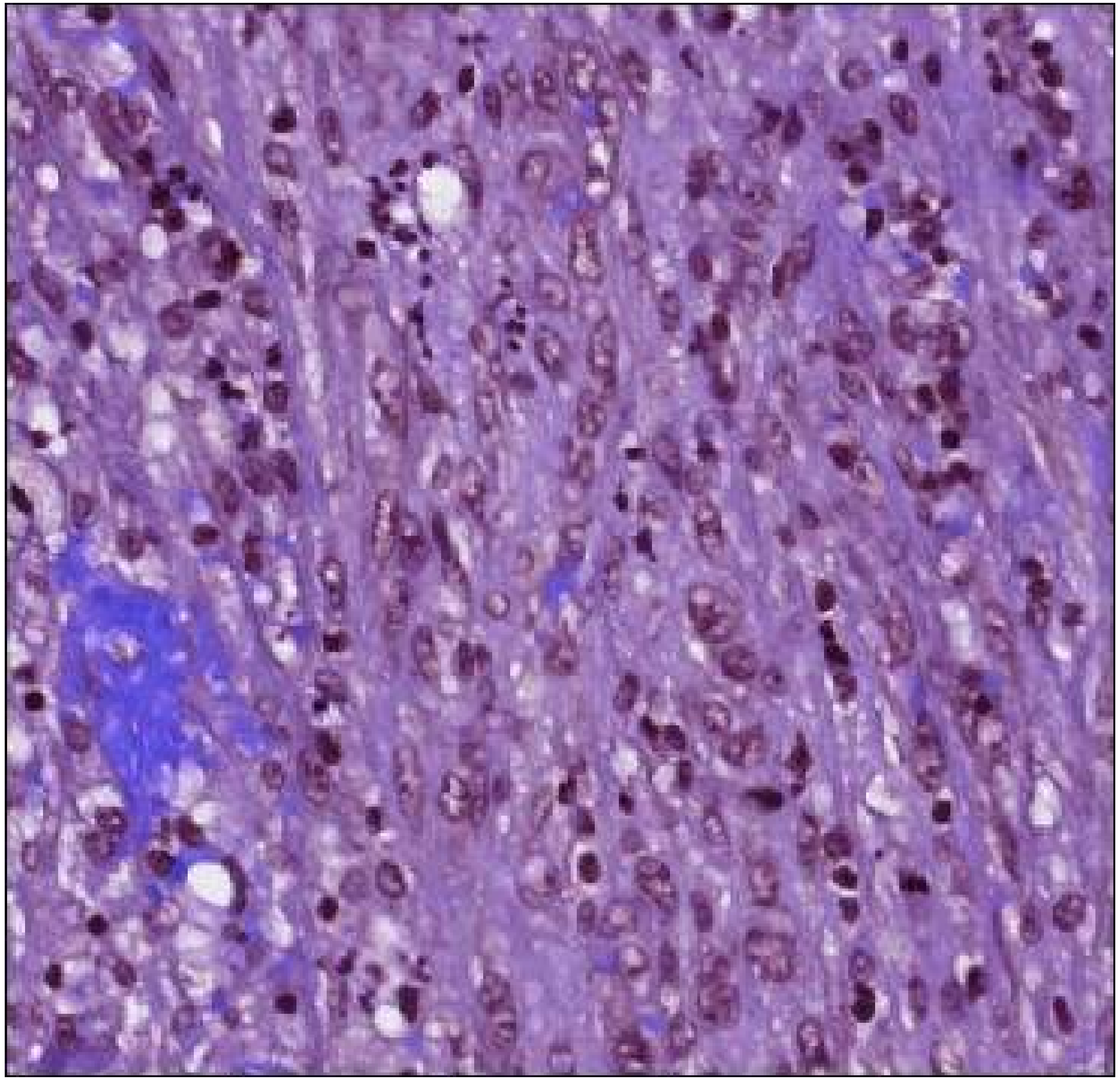
Preprocessing - Approaches in Literature

- For Color-to-Grayscale, 13 mainly used methods. In terms of image recognition, best two are the simplest ones (Kanan, Garrison 2):
  - «Intensity» : 1/3*(R+G+B)
  - «Gleam» : 1/3*(R’+G’+B’) – (Same with intensity, gamma correction added)

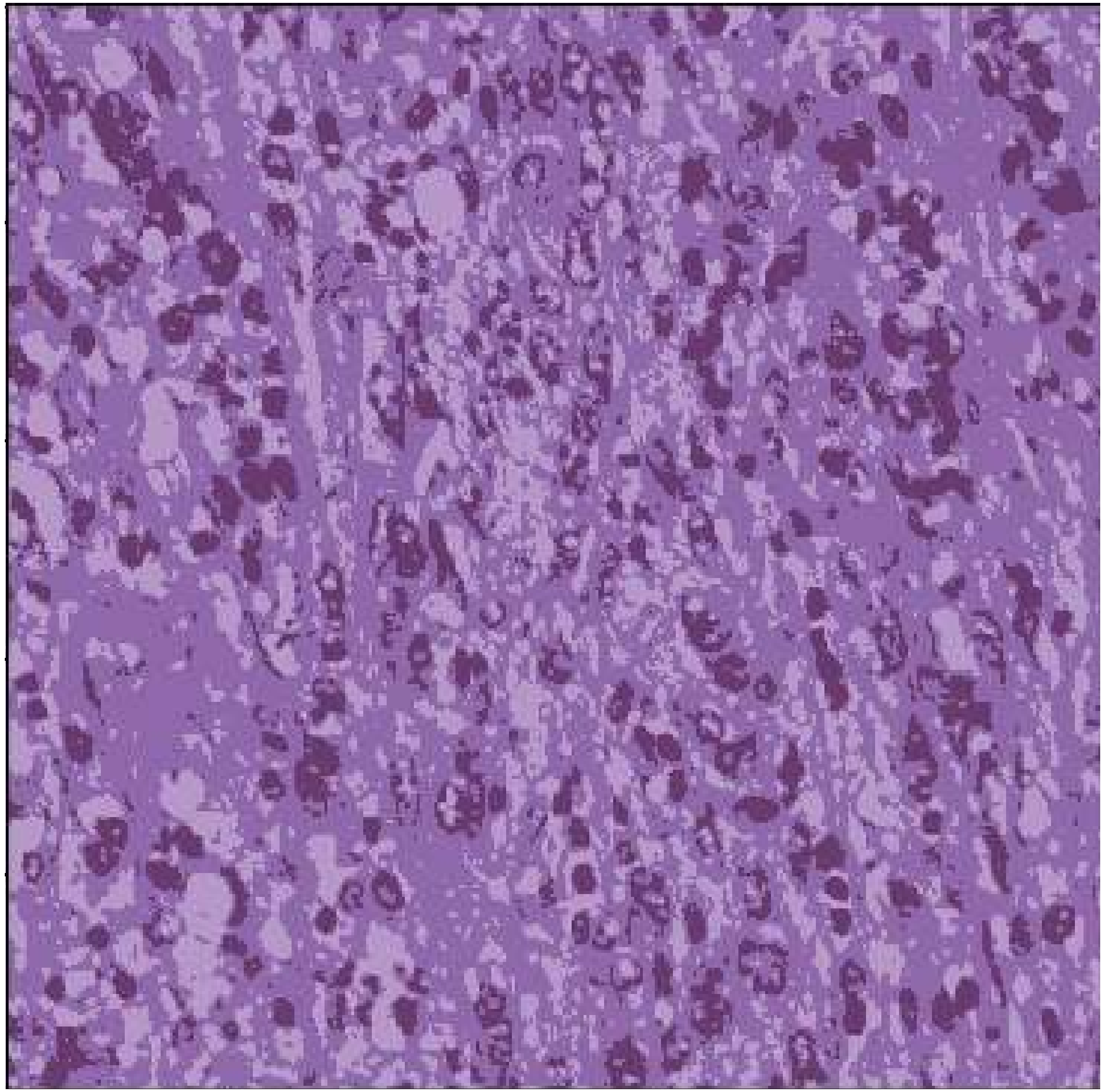
- In cell extraction, similar approaches are used. Choose of «Value» from HSV space is one of them, which is criticized in the paper above. However, there isn’t an all-purpose method such as the one proposed here for this non-trivial task (Pinidiyaarachchi).

Preprocessing

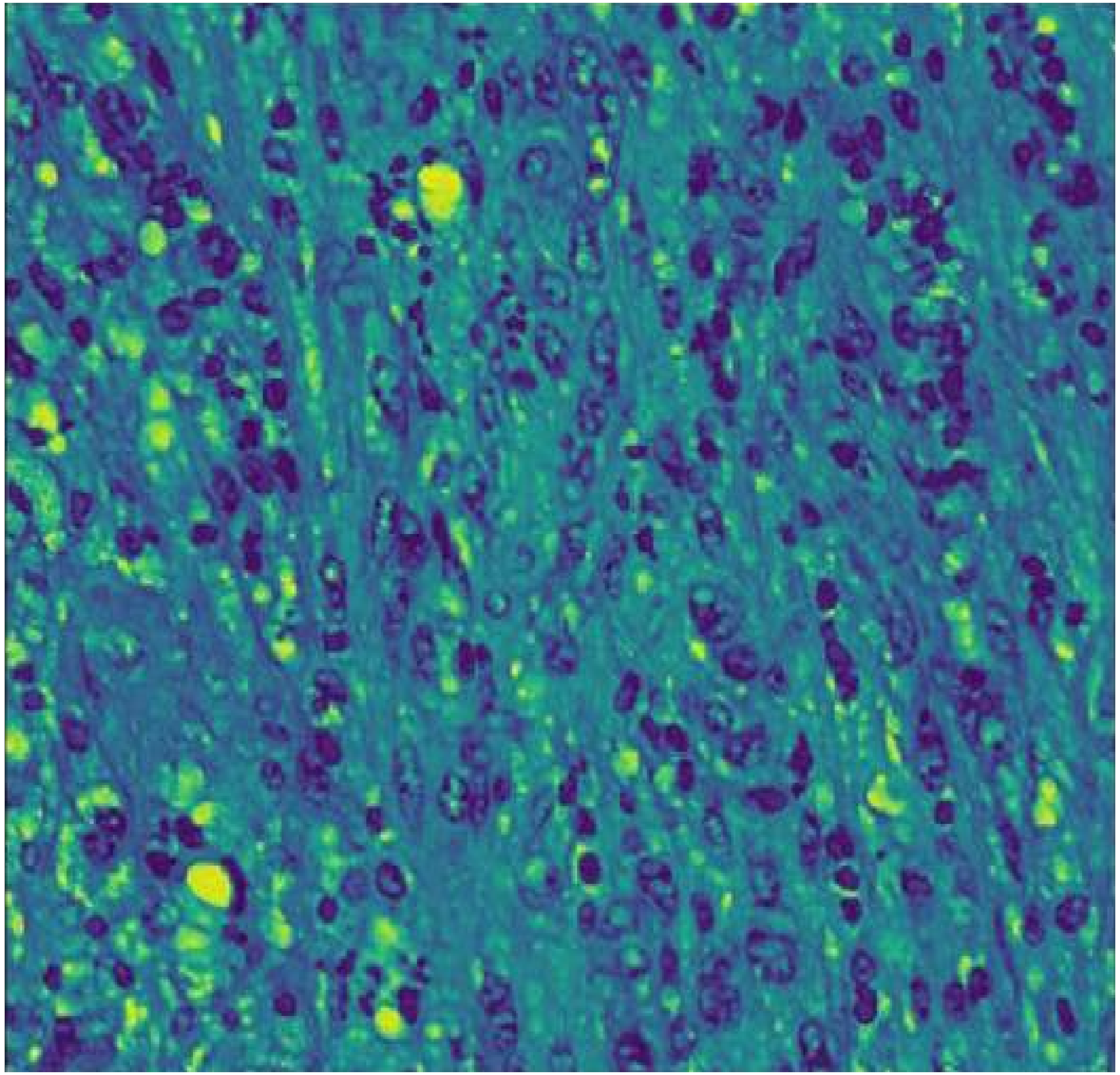
- Convert 3D to 2D – with low intensity as the target
  - ✓ Apply K-means clustering with K=3 (Background, cytoplasm and nuclei)
  - ✓ Choose the nuclei center by minimizing cell score : area / circularity = perimeter^2
  - ✓ Normalize the given center by the background values, which maximizes the cell score above.
  - ✓ Calculate the Euclidian distance to the normalized center and map to (0-255)



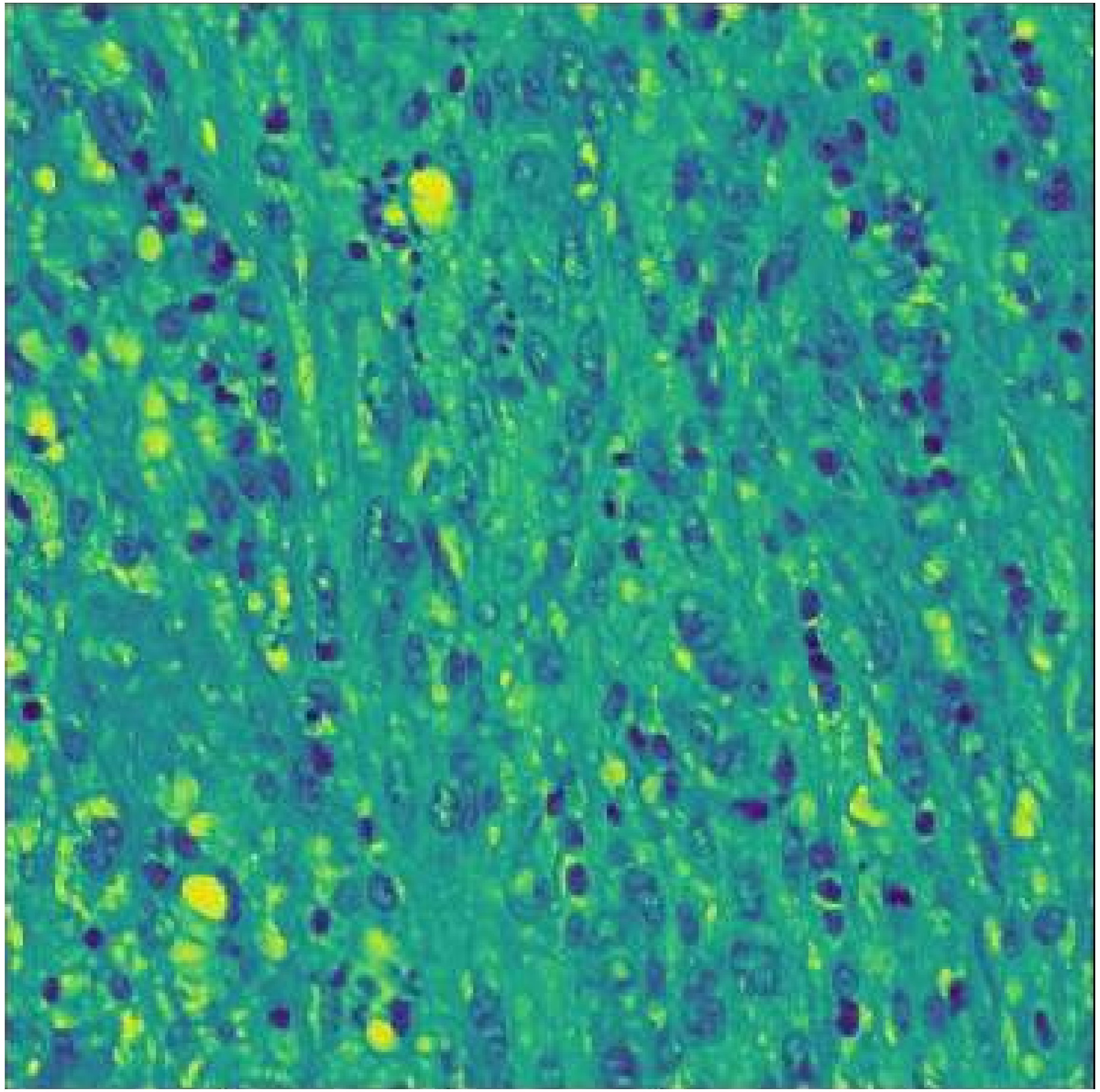
Original Image



K-Means Clustering (K=3)



Preprocessed Image



Preprocessed Image with «Intensity» Method

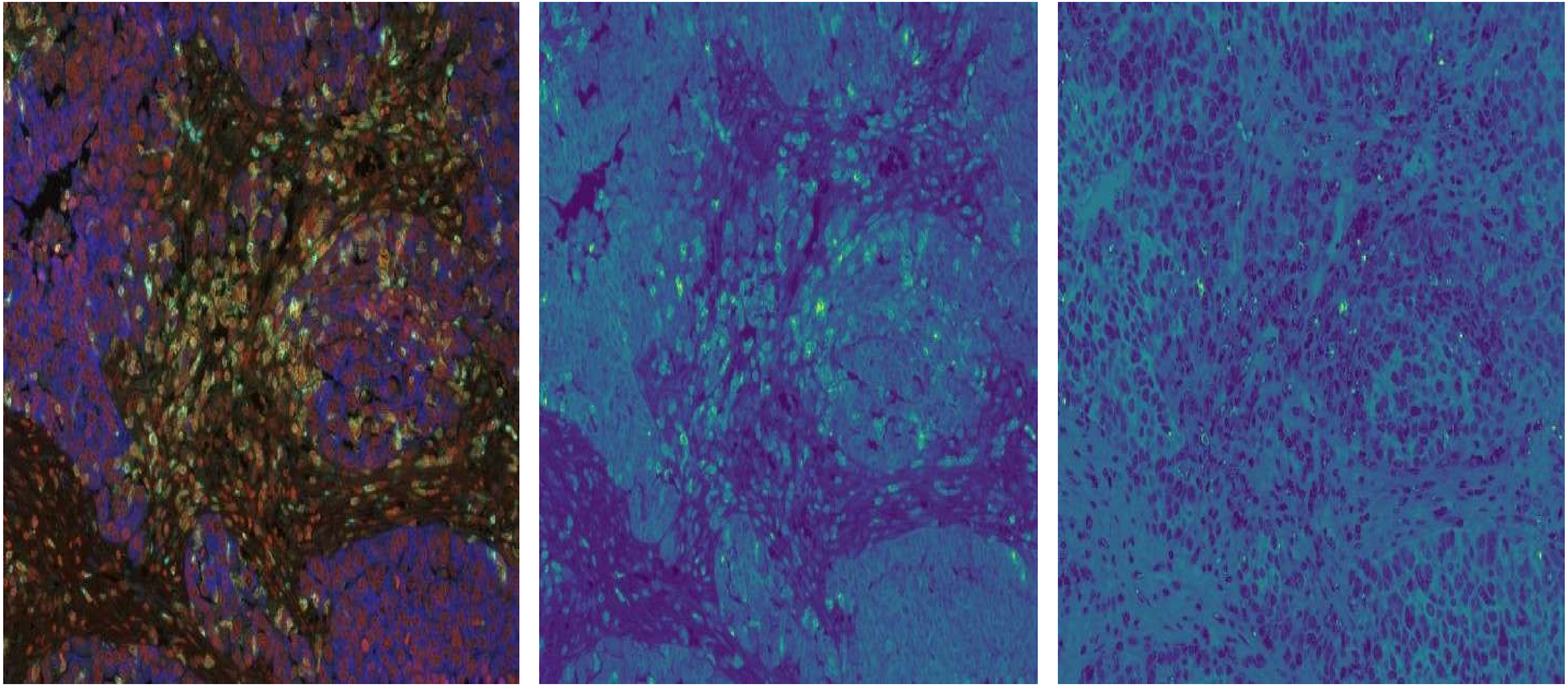


Figure 7-9 - From left to right: Initial image with multicolored staining, <Intensity> method, Proposed method

Advantages - Disadvantages

- Applicable to nearly any type of staining
- Output is compatible with segmentation algorithms
- Disregards the irregular stains on the tissue

- Maxima-minima points are not necessarily preserved, further smoothing is required.

Segmentation - Approaches in Literature

- Tens of other methods, generally uses:
  - Watershed algorithm
  - Edge detection
  - Neural Networks – ML
  - Thresholding [3]

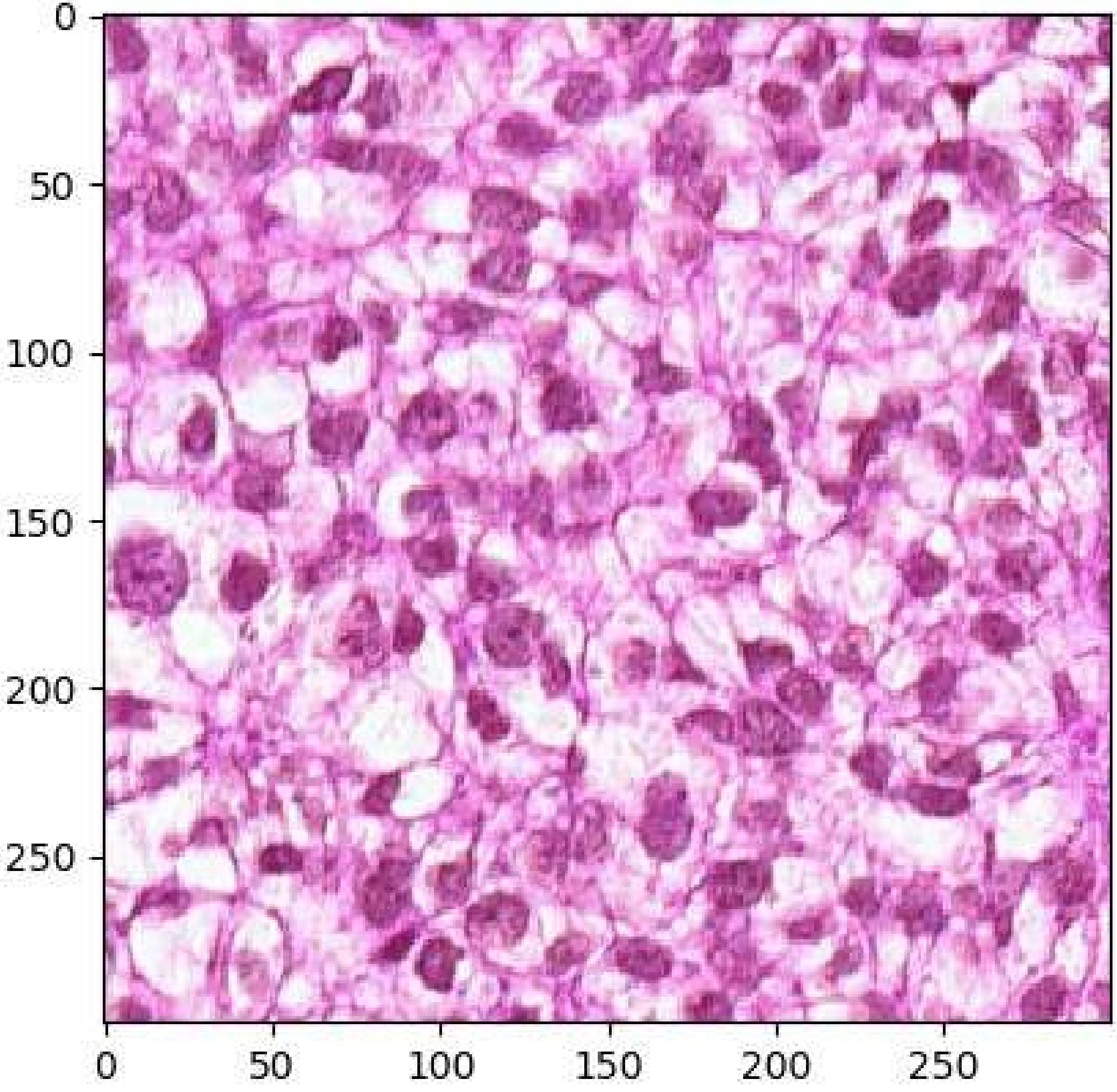
- Watershed is generally used after the preprocessing methods with distance mapping calculations. 12 papers used watershed or region growing among 17 papers presented by Xing et al.

- However, usual watershed tend to create over-segmentated regions because of local minimas. We need an approach that can combine minimas if necessary.

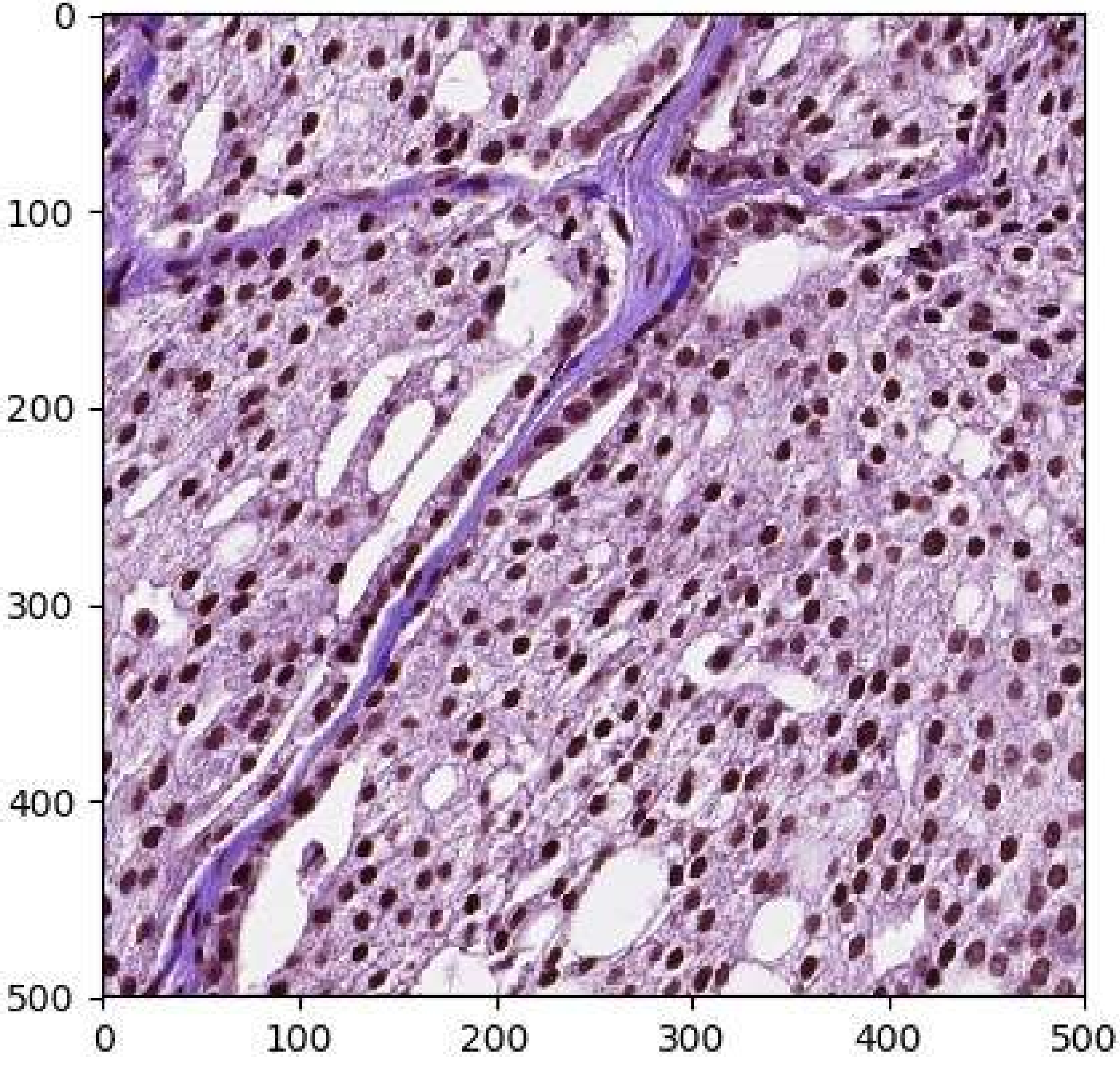
Cluster Segmentation

- Segment the image into clusters as much as possible, while gaining information on the expected cell shape of the given tissue

  - ✓ Smooth the image and apply histogram equalization
  - ✓ Apply K-means clustering for ideally K>10, select the darkest layer as the initial seeds and sapply watershed one layer after another.
  - ✓ If a seed grows over another one with lower priority during this process, lower priority seed is discarded.



Original Image



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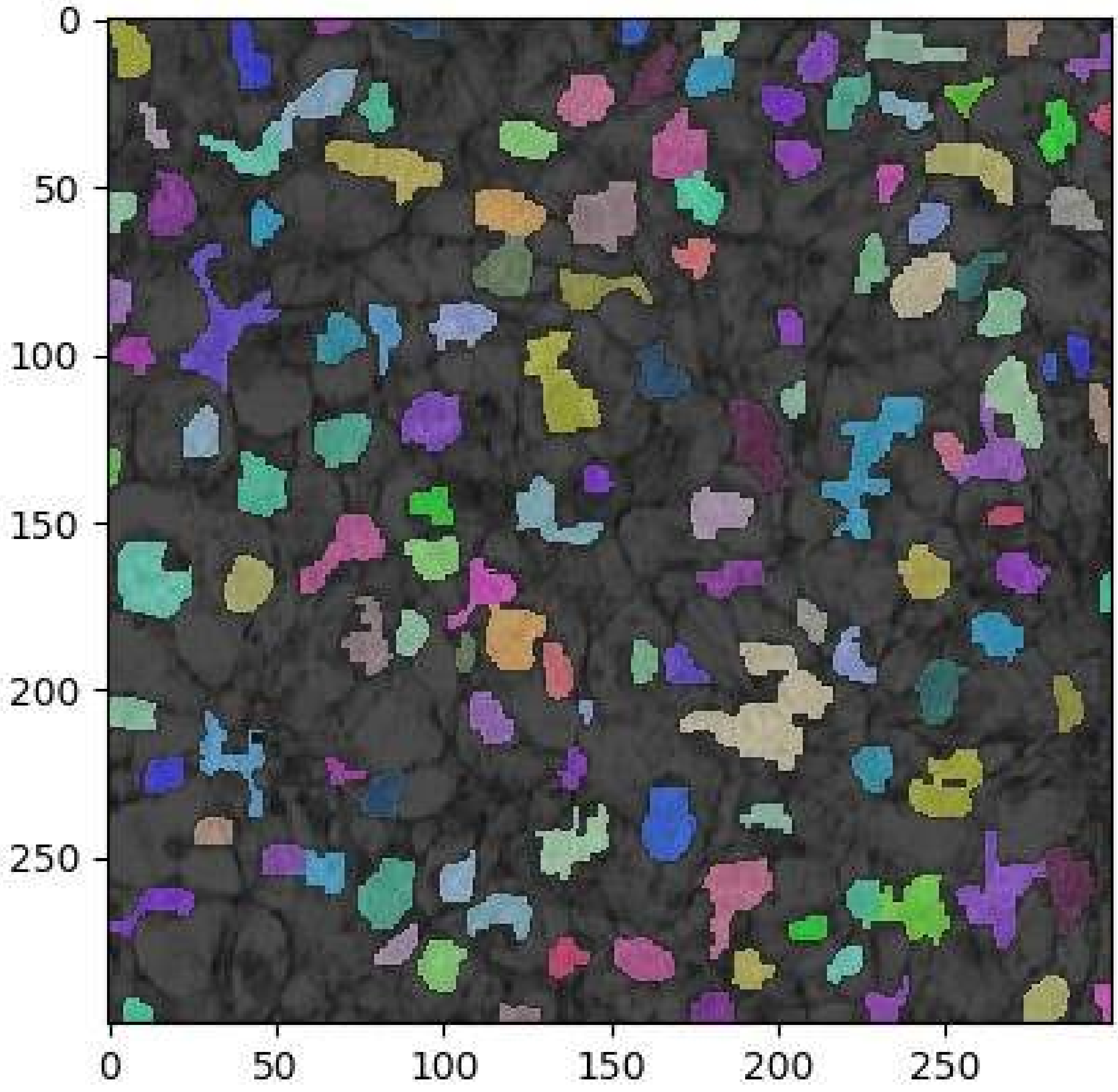
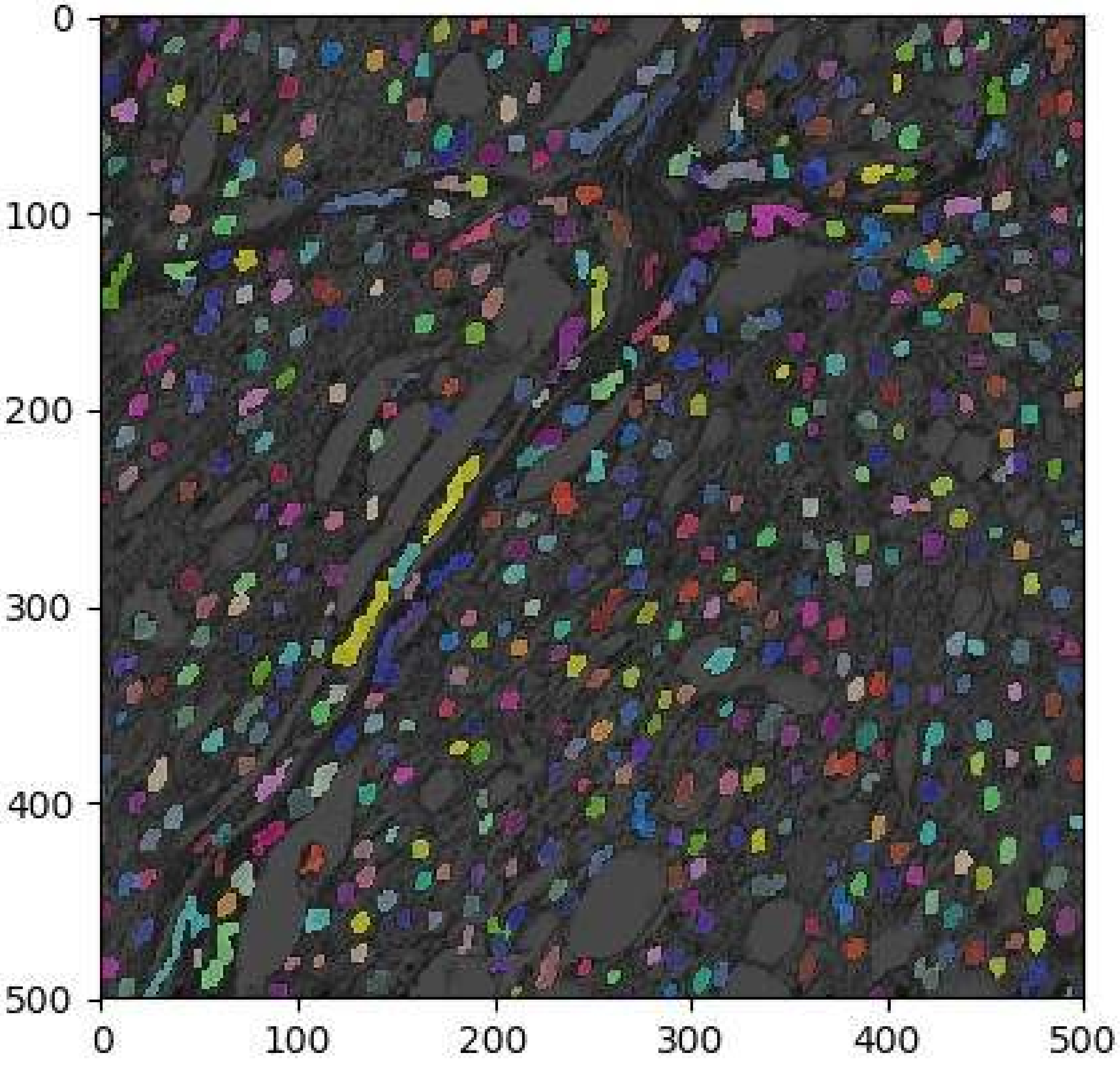


Segmented Image (K=10)



Segmented Image (K=10)

Advantages - Disadvantages

- We have a degree of certainity for cells, since there is a priority relationship between the layers and higher layers have better probability of being cells. This can be used to determine a quality threshold, or even create a supervised learning model.
- For relax stopping points, this algorithm misses little to no cells.
- Does not divide any elongated nuclei.

- No robust stopping time yet!
- Can’t segmentate too close nucleis – needs a follow up method
- Has some problems with non-homogenous cell staining

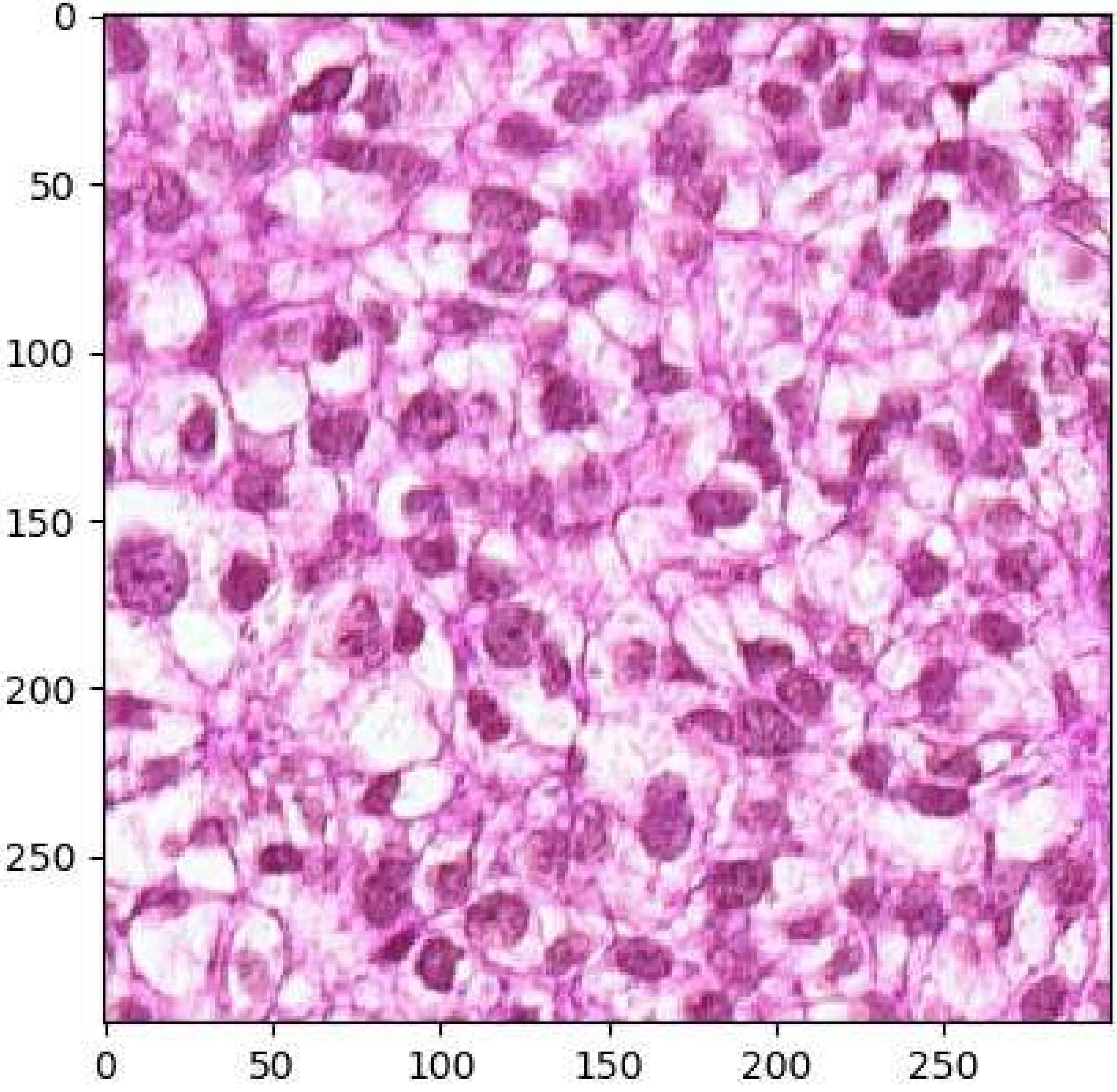
Convex Cutting - Approaches in Literature

- The use of convexity defects is a frequently utilized approach, along with ellipse fitting (Indumathi et al.; Kothari et al.; Wang et al.; Wienert et al.; Xing et al.).
- Normally, after watershed-like approaches, region merging is a often used alternative, since watershed tends to over-segmentate.
- However, for under-segmentation such as this, generally convexity cutting and ellipse fitting methods are used.

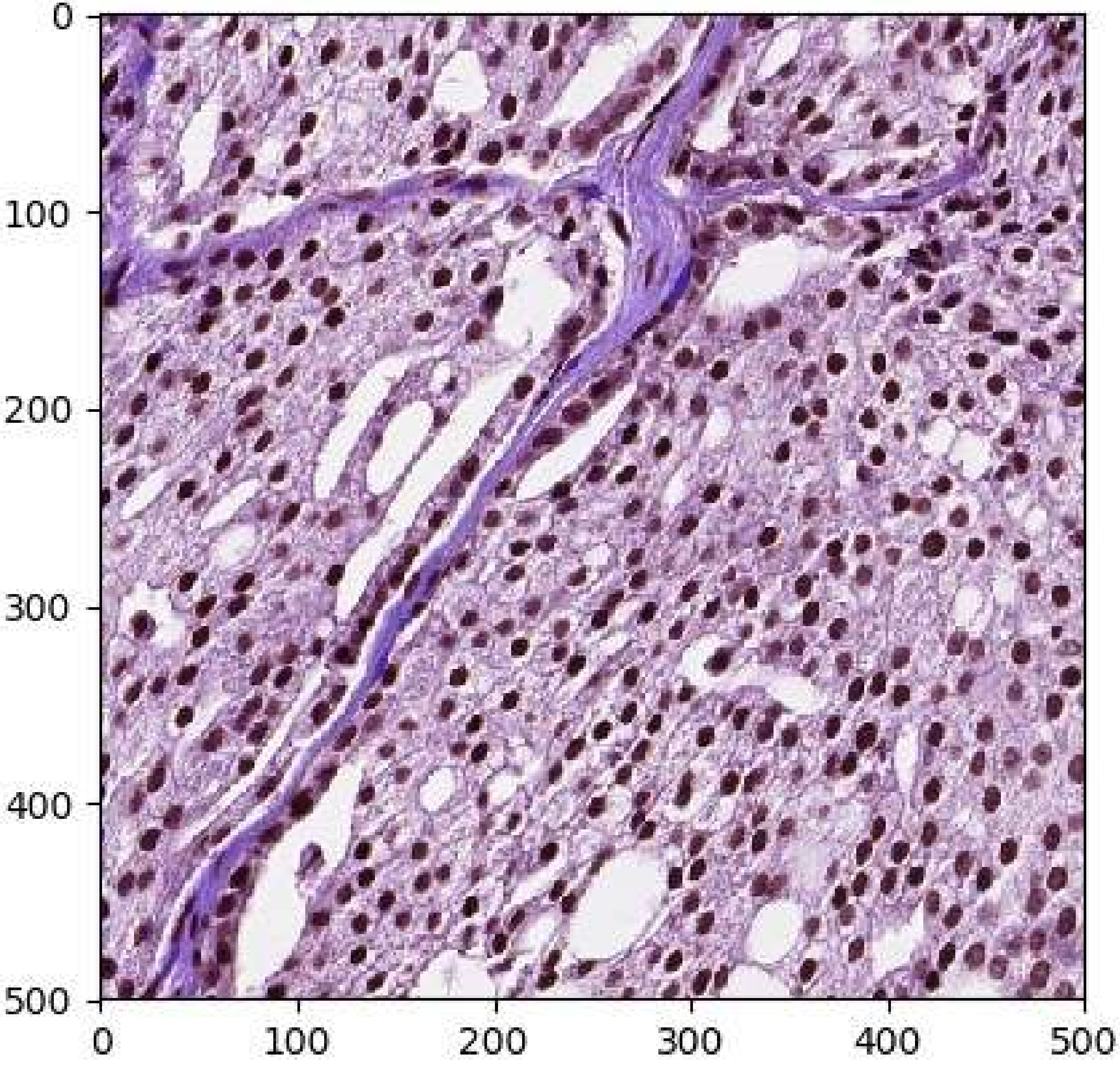
Cluster Cutting

- Cut the given clusters by lines originating from their convexity defects.

  - ✓Calculate a score for each region (perimeter^2); and threshold them using the layer with highest priority from the previous algorithm.
  - ✓To the clusters that are above the given threshold, apply the cutting algorithm.
  - ✓Designate a depth threshold for defects to cut (e.g. 3 pixels).
  - ✓Calculate a score for each possible line passing from two defects. Score is calculated according to the angle of the possible line and depth of given defects, described by Wienert et al.
  - ✓Cut them recursively until they become a suitable cell by score, both resulting regions are smaller than the possible cell size or there is no more defect to cut.



Original Image



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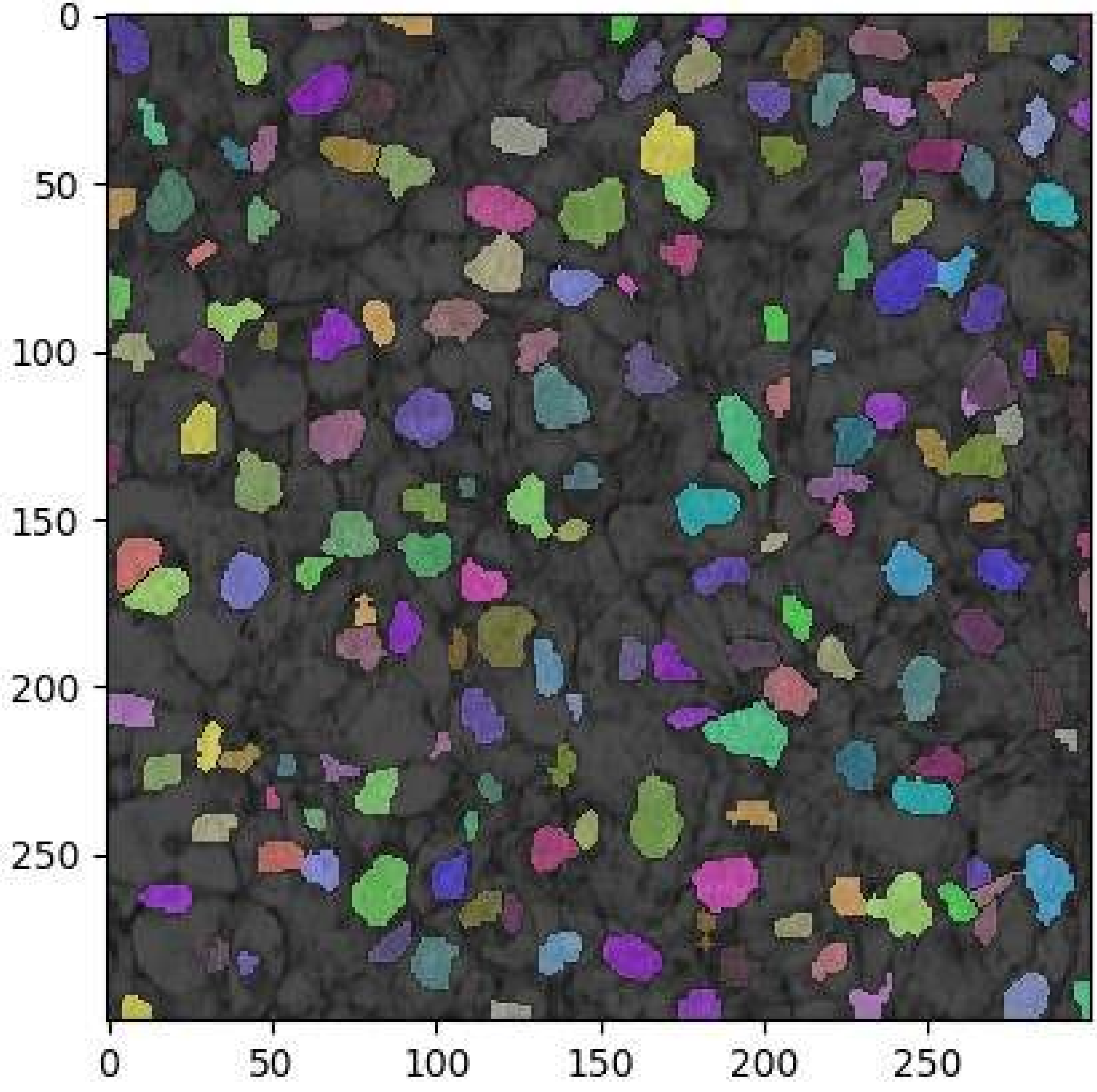
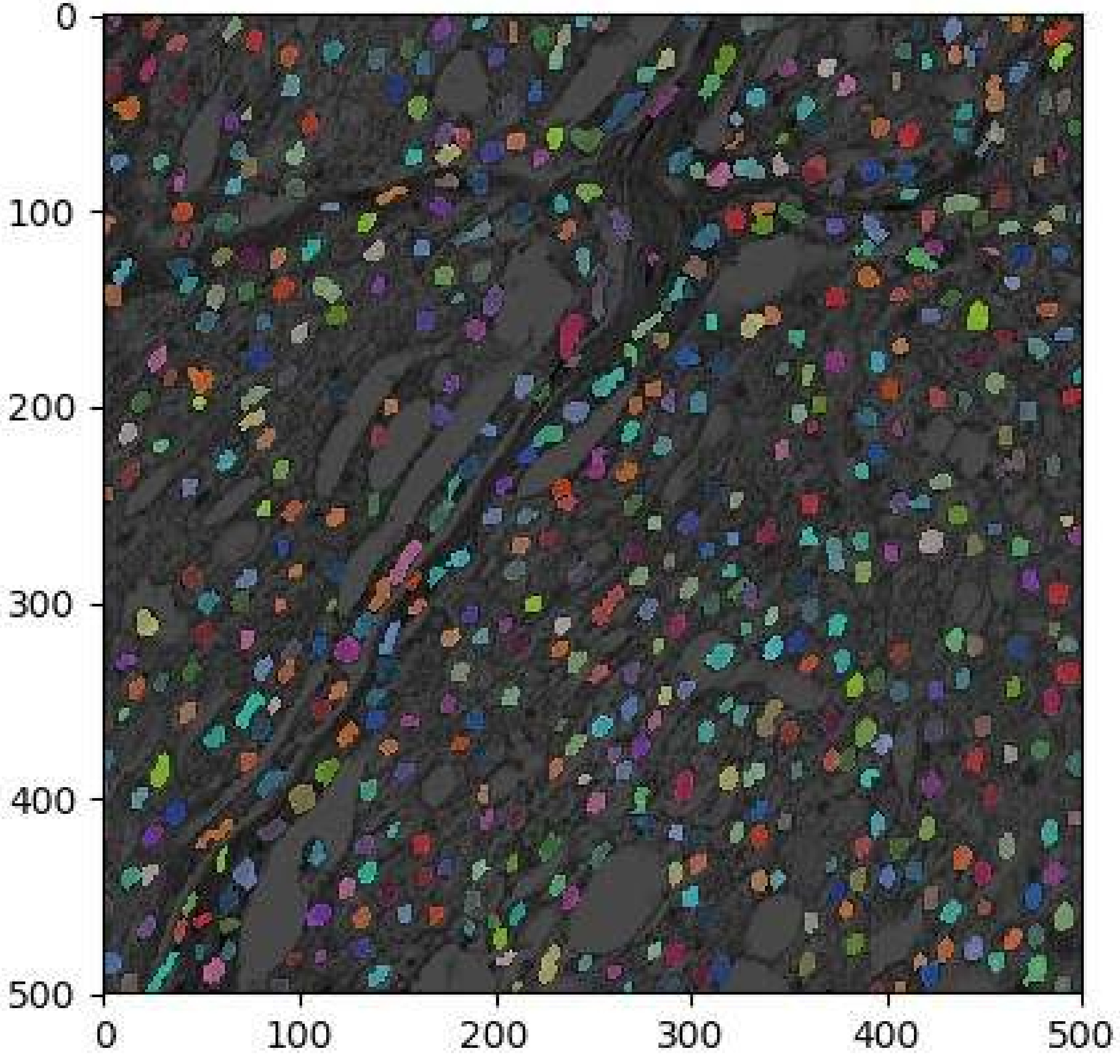
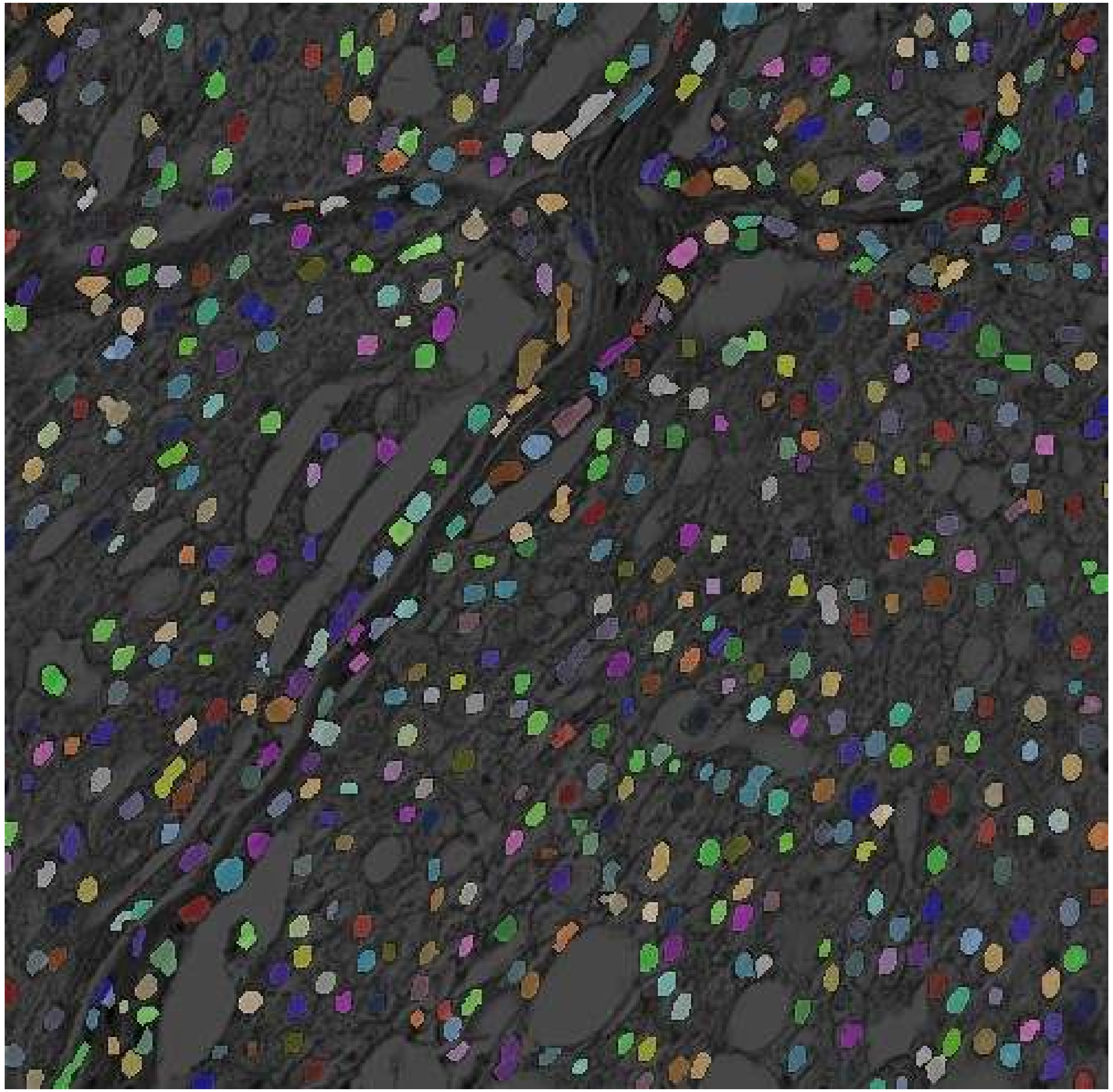


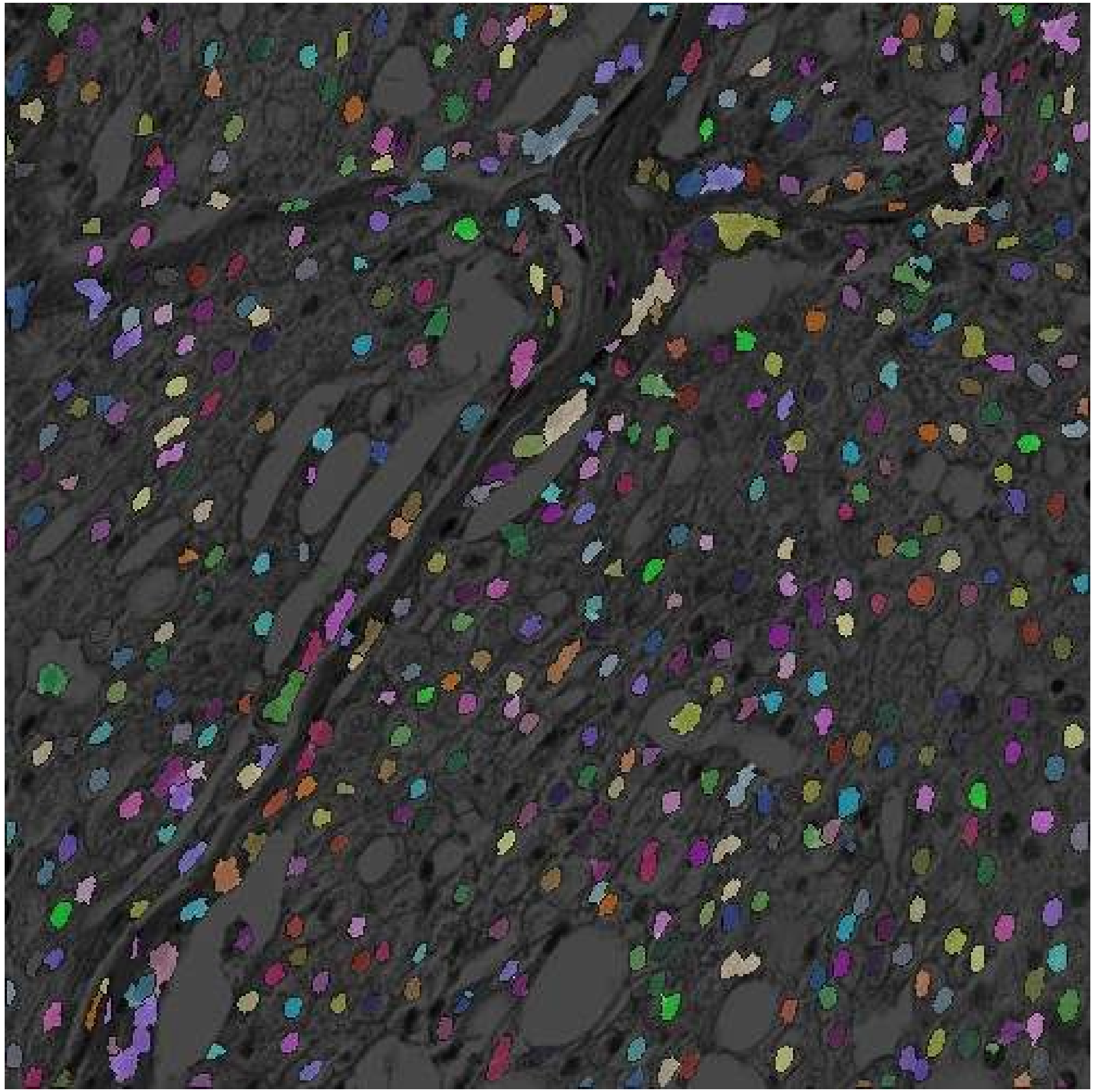
Cut Image (depth threshold = 2.5)



Cut Image (depth threshold = 2.5)

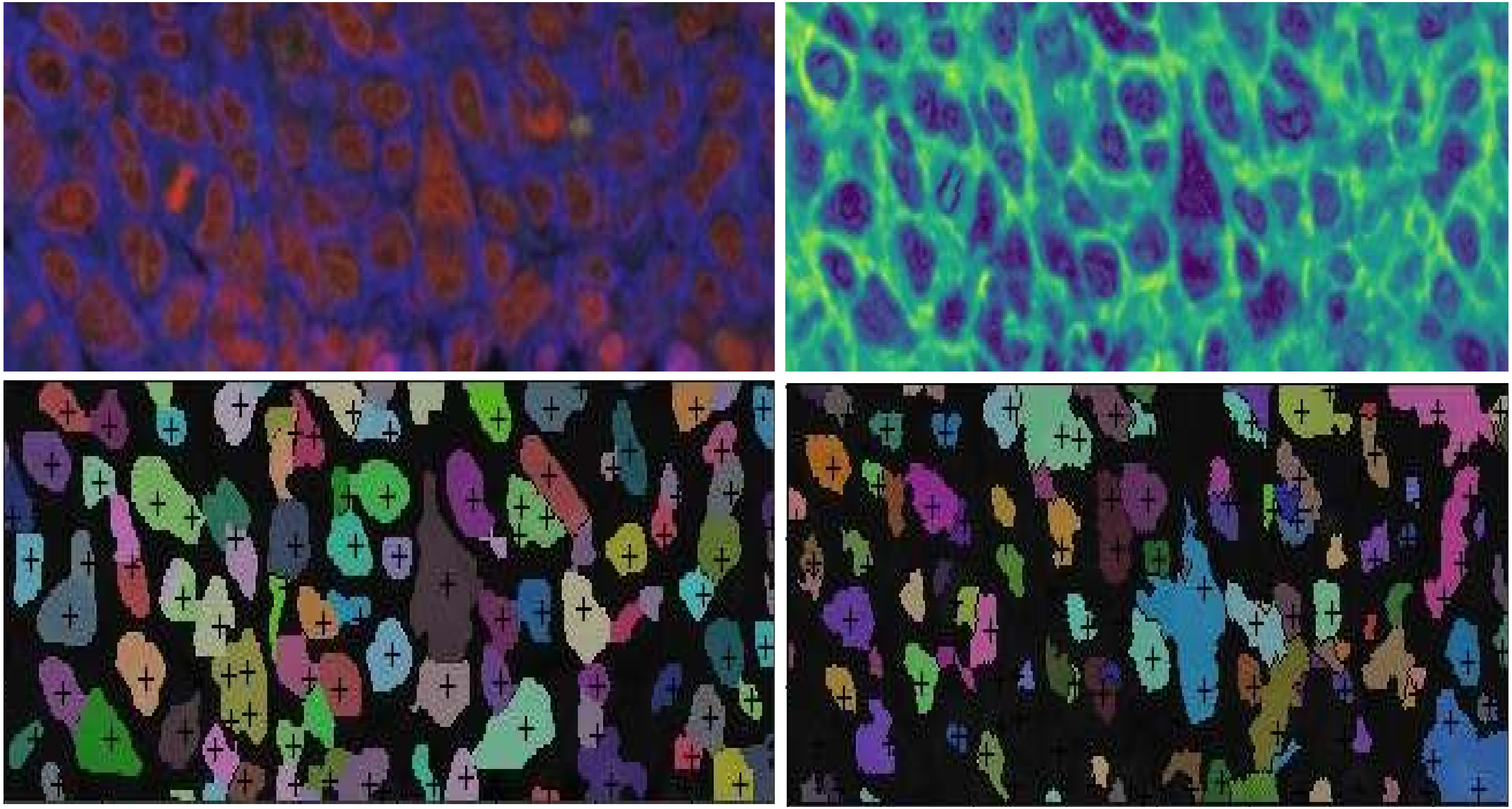


Proposed Method



State of the art watershed algorithm

(with distance mapping)



From left to right, top to bottom: Initial image, Preprocessed image, Processed Image, State of the art watershed algorithm with distance mapping

Advantages - Disadvantages

- Two-cell and three-cell clusters can easily be separated without any side effect.
- Cells with deformed or too largely segmented regions are safely shaved.
- Algorithm has a clearly defined stopping point.

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- Clusters with more than three cells are hard to divide, and even when the division is correct in terms of cell count, the shapes are often wrong.
- Algorithm does not have a specific depth threshold for convexity defects.

Future Work

- Cell cutting algorithm lacks the measures that segmentation algorithm uses to prevent cell nuclei divisions. As a result, some of the cuts are done through prioritized areas. This issue can be prevented by simple algorithms to penalize any such cuts using the information from the segmentation method, yet an alternative approach is needed in its place which may allow parabolic or hyperbolic lines to divide the aforementioned clusters.
- Prioritized data from the segmentation algorithm can be used to train supervised networks, which may allow automatic detection and division of badly segmented cells.
- Preprocessing method may be utilized for different areas of image processing, and needs to be tested for its performance on image analysis datasets.

References

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